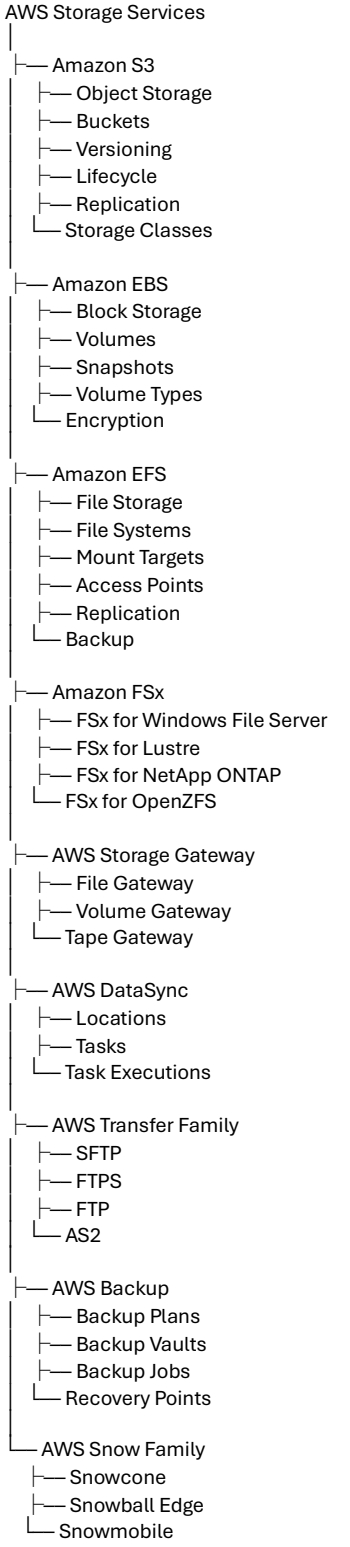
Amazon AWS Storage Services — Console Dashboard Wise Important Interview Questions & Answers

AWS Storage Services Console Map



---

1. What are AWS Storage Services?
Answer:
AWS Storage Services are services used to store, protect, transfer, and manage data in AWS. Different services are designed for object, block, file, hybrid, and backup storage requirements. We choose the service based on how the application needs to access the data.
For example:

Suppose our application needs to store images, attach a disk to EC2, and share files between multiple servers.
We can use **S3 for objects, EBS for EC2 disks, and EFS for shared files**.
Example:
Application
│
├── Images / Videos → S3
│
├── EC2 Disk → EBS
│
└── Shared Files → EFS
Simple ga gurthupettuko:
AWS Storage = Object + Block + File + Backup + Transfer

---

2. What are the main types of storage in AWS?
Answer:
The three major storage types are object storage, block storage, and file storage. S3 provides object storage, EBS provides block storage, and EFS provides file storage. Each type works differently depending on the application's access requirement.
For example:
If we store an image as an object, we use S3.
If EC2 needs a disk like a hard drive, we use EBS.
If multiple EC2 servers need the same shared files, we can use EFS.
Example:
Object → S3
Block → EBS
File → EFS
Simple ga gurthupettuko:
S3 = Object | EBS = Block | EFS = File

---

3. What is Amazon S3 used for?
Answer:
Amazon S3 is an object storage service used to store files and data as objects inside buckets. It is commonly used for images, videos, backups, logs, documents, and static website content. S3 is designed for highly durable and scalable storage.
For example:
Suppose users upload profile pictures to our application.
Instead of storing them on the EC2 server, we store the images in an S3 bucket.
Example:
User
↓
Application
↓
S3 Bucket
↓
Profile Images
Simple ga gurthupettuko:
S3 = Store objects/files

---

4. What is Amazon EBS used for?
Answer:
Amazon EBS provides block storage that can be attached to EC2 instances. It works like a virtual hard disk for an EC2 server. Different EBS volume types are available for different performance and workload requirements.
For example:
Suppose an EC2 server needs a disk to install an operating system and store application data.
We create an EBS volume and attach it to the EC2 instance.
Example:
EC2
↓
EBS Volume
↓
Application Data
Simple ga gurthupettuko:
EBS = Disk for EC2

---

5. What is Amazon EFS used for?
Answer:
Amazon EFS is a managed file storage service that can be mounted by multiple EC2 instances. It provides a shared file system that multiple servers can access at the same time. It is useful when applications need common files across multiple servers.
For example:
Suppose we have several EC2 servers running the same application.
If all servers need access to the same files, we can mount one EFS file system to those servers.
Example:
EC2-1 ──┐

EC2-2 ──┼── EFS
EC2-3 ──┘
↓
Shared Files
Simple ga gurthupettuko:
EFS = Shared file system

---

6. What is Amazon FSx?
Answer:
Amazon FSx provides managed file systems for specific workloads and operating systems. It supports file systems such as Windows File Server, Lustre, NetApp ONTAP, and OpenZFS. It is useful when an application needs features or performance provided by a particular file system.
For example:
Suppose a company has Windows applications that require Windows file shares.
We can use **FSx for Windows File Server** instead of building and maintaining our own Windows file server.
Example:
Application
↓
Amazon FSx
↓
Managed File System
Simple ga gurthupettuko:
FSx = Managed specialized file systems

---

7. What is the difference between EFS and FSx?
Answer:
EFS is a general-purpose managed file system commonly used with Linux workloads and shared file access. FSx provides managed file systems designed for specific technologies such as Windows, Lustre, NetApp ONTAP, and OpenZFS. We choose based on the application's operating system and file-system requirements.
For example:
For a Linux application needing a simple shared file system, we may choose EFS.
For a Windows application requiring SMB file shares, we may choose FSx for Windows File Server.
Example:
Linux Shared Files
↓
EFS

Windows / Specialized
↓
FSx
Simple ga gurthupettuko:
EFS = General shared file storage | FSx = Specialized file systems

---

8. What is AWS Storage Gateway?
Answer:
AWS Storage Gateway is a hybrid storage service that connects on-premises environments with AWS storage. It allows applications running on-premises to use AWS storage without completely moving the application immediately. It provides different gateway types for different storage requirements.
For example:
Suppose a company has servers in its data center but wants to store data in AWS.
Storage Gateway can connect the on-premises environment with AWS storage.
Example:
On-Premises
↓
Storage Gateway
↓
AWS Storage
Simple ga gurthupettuko:
Storage Gateway = On-premises ↔ AWS storage

---

9. What are the main types of Storage Gateway?
Answer:
Storage Gateway provides File Gateway, Volume Gateway, and Tape Gateway. File Gateway provides file access to S3, Volume Gateway provides block storage integration, and Tape Gateway provides virtual tape functionality. The correct gateway depends on the existing application requirement.
For example:
If an on-premises application needs file access backed by S3, we can use File Gateway.
If a backup application expects virtual tapes, we can use Tape Gateway.
Example:
Storage Gateway
│

├— File Gateway
├— Volume Gateway
└— Tape Gateway

Simple ga gurthupettuko:
File = File Gateway | Block = Volume Gateway | Tape = Tape Gateway

---

10. What is AWS DataSync?
Answer:
AWS DataSync is a data transfer service used to move data between on-premises storage and AWS storage services. It can automate and accelerate large-scale data transfers. It is commonly used for migration, backup, and synchronization workloads.
For example:
Suppose a company has millions of files on an on-premises file server.
DataSync can transfer those files to AWS storage such as S3 or EFS.
Example:
On-Premises Storage
↓
DataSync
↓
AWS Storage
Simple ga gurthupettuko:
DataSync = Move data to/from AWS

---

11. What is an AWS DataSync Location?
Answer:
A DataSync Location represents the source or destination storage system used in a data transfer task. It can represent storage such as an on-premises file system or an AWS storage service. DataSync uses these locations to know where data should come from and where it should go.
For example:
Suppose we want to move files from an on-premises NFS server to EFS.
We create one location for the NFS server and another location for EFS.
Example:
NFS Location
↓
DataSync
↓
EFS Location
Simple ga gurthupettuko:
Location = Source or destination

---

12. What is a DataSync Task?
Answer:
A DataSync Task defines how data should be transferred between two locations. It contains the source, destination, and transfer configuration. We can run the task whenever we need to transfer or synchronize data.
For example:
Suppose our source is an on-premises NFS server and our destination is an S3 bucket.
We create a DataSync task connecting these two locations.
Example:
Source Location
↓
DataSync Task
↓
Destination Location
Simple ga gurthupettuko:
Task = Rules for data transfer

---

13. What is AWS Transfer Family?
Answer:
AWS Transfer Family provides managed file-transfer services using protocols such as SFTP, FTPS, FTP, and AS2. It allows external users or applications to transfer files into and out of AWS storage. It is useful when users already depend on standard file-transfer protocols.
For example:
Suppose a partner company already uses SFTP to send files.
Instead of creating our own SFTP server, we can use AWS Transfer Family with S3 or EFS.
Example:
Partner
↓
SFTP
↓
Transfer Family
↓
S3 / EFS

Simple ga gurthupettuko:
Transfer Family = Managed SFTP/FTP file transfer

---

14. What storage can AWS Transfer Family use?
Answer:
AWS Transfer Family can integrate with AWS storage such as Amazon S3 and Amazon EFS depending on the configuration. This allows users to transfer files using supported protocols while the data is stored in AWS storage. The transfer server handles the protocol connection while the storage service holds the data.
For example:
Suppose a partner uploads a file through SFTP.
Transfer Family can store that file in an S3 bucket.
Example:
Partner
↓
SFTP
↓
Transfer Family
↓
S3 Bucket
↓
File
Simple ga gurthupettuko:
Transfer Family = File transfer layer

---

15. What is AWS Backup?
Answer:
AWS Backup is a centralized service used to configure and manage backups across supported AWS services. It allows us to create backup plans, backup schedules, retention rules, and backup vaults. This makes backup management easier across multiple AWS resources.
For example:
Suppose our application uses EBS, RDS, and EFS.
Instead of managing backups separately, we can use AWS Backup to centrally manage supported backups.
Example:
EBS ─┐
RDS ─┼─ AWS Backup
EFS ─┘ ↓
Backup Vault
Simple ga gurthupettuko:
AWS Backup = Centralized backup management

---

16. What is a Backup Plan in AWS Backup?
Answer:
A Backup Plan defines when backups should run and how long those backups should be retained. It can contain backup rules with schedules, lifecycle settings, and backup vault destinations. Resources can be assigned to the plan using resource assignments.
For example:
Suppose we want our application resources to be backed up every day and retained for 30 days.
We can create a Backup Plan with that schedule and retention requirement.
Example:
Backup Plan
↓
Daily Backup
↓
Retention
↓
Backup Vault
Simple ga gurthupettuko:
Backup Plan = When + How long to keep backups

---

17. What is a Backup Vault?
Answer:
A Backup Vault is a logical container where AWS Backup recovery points are stored. It helps organize and control access to backups. We can also use vault-level features such as encryption and access policies.
For example:
Suppose our production backups need to be stored separately from development backups.
We can create different backup vaults for different backup requirements.
Example:
AWS Backup
↓
Backup Vault
↓
Recovery Points

Simple ga gurthupettuko:
Backup Vault = Container for backups

---

18. What is a Recovery Point?
Answer:
A recovery point is a point-in-time backup that can be used to restore a supported AWS resource. AWS Backup creates recovery points when backup jobs successfully complete. Recovery points are stored inside backup vaults.
For example:
Suppose an EC2 application's EBS volume is backed up every night.
Each successful backup creates a recovery point that can later be used for recovery.
Example:
Resource
↓
Backup Job
↓
Recovery Point
↓
Backup Vault
Simple ga gurthupettuko:
Recovery Point = Backup copy used for recovery

---

19. What is the AWS Snow Family?
Answer:
AWS Snow Family provides physical devices used to move or process large amounts of data when transferring data over the network is difficult or too slow. These devices can be shipped between the customer and AWS. They are useful for large-scale data migration and edge workloads.
For example:
Suppose a company has hundreds of terabytes of data and its internet connection is too slow for practical transfer.
It can use a Snow Family device to physically move the data to AWS.
Example:
On-Premises Data
↓
Snow Device
↓
Ship to AWS
↓
AWS
Simple ga gurthupettuko:
Snow Family = Physical data transfer to AWS

---

20. What is Snowball Edge?
Answer:
Snowball Edge is a physical AWS device used for large-scale data transfer and certain edge computing workloads. It can store significant amounts of data and can process workloads locally. It is useful when network transfer is not practical or when processing must happen near the data source.
For example:
Suppose a remote location has a large amount of data but poor internet connectivity.
A Snowball Edge device can collect the data locally and later transfer it to AWS.
Example:
Remote Location
↓
Snowball Edge
↓
Physical Transfer
↓
AWS
Simple ga gurthupettuko:
Snowball Edge = Physical data transfer + Edge processing

---

21. What is the difference between S3, EBS, and EFS?
Answer:
S3 is object storage, EBS is block storage, and EFS is file storage. S3 is accessed using objects and APIs, EBS behaves like a disk attached to EC2, and EFS provides a shared file system. The choice depends on how the application needs to access the data.
For example:
For application images, we can use S3.
For an EC2 operating-system disk, we use EBS, and for shared files across EC2 servers, we use EFS.
Example:
Images / Backups → S3
EC2 Disk → EBS
Shared Files → EFS

Simple ga gurthupettuko:
S3 = Object | EBS = Block | EFS = File

---

22. Which AWS storage service should you use for EC2 root storage?
Answer:
Amazon EBS is commonly used for EC2 root storage. The root volume contains the operating system and is attached to the EC2 instance as a block device. The EBS volume can be managed separately from the EC2 instance.
For example:
Suppose we launch an EC2 instance with Amazon Linux.
The operating system is installed on the instance's EBS root volume.
Example:
EC2 Instance
↓
Root EBS Volume
↓
Operating System
Simple ga gurthupettuko:
EC2 Root Disk = EBS

---

23. Which AWS storage service should you use for shared files between multiple EC2 instances?
Answer:
Amazon EFS is a common choice when multiple Linux EC2 instances need to access the same shared file system. Multiple instances can mount the same EFS file system. This is useful for shared application files and common directories.
For example:
Suppose three EC2 servers are running the same application and all need access to uploaded files.
We can mount the same EFS file system on all three servers.
Example:
EC2-1 ─┐
EC2-2 ─┼─ EFS
EC2-3 ─┘
↓
Shared Files
Simple ga gurthupettuko:
Multiple EC2 + Same Files = EFS

---

24. Which storage service should you use for backups and archival data?
Answer:
Amazon S3 is commonly used for backups, archival data, logs, documents, and other objects that need scalable storage. S3 also provides different storage classes for different access patterns and costs. Lifecycle policies can automatically move old data to more appropriate storage classes.
For example:
Suppose our application creates database backups every day.
We can store the backup files in S3 and move older backups to lower-cost storage classes using lifecycle rules.
Example:
Application
↓
Backup
↓
S3
↓
Lifecycle
↓
Lower-Cost Storage
Simple ga gurthupettuko:
Backup / Archive = S3

---

25. Which AWS service would you choose for Windows file shares?
Answer:
Amazon FSx for Windows File Server is designed for Windows-based file storage and SMB file shares. It provides a managed Windows file system without requiring us to maintain the underlying file servers ourselves. It is useful for Windows applications that depend on traditional file shares.
For example:
Suppose a Windows application expects a shared folder using SMB.
Instead of managing our own Windows file server, we can use FSx for Windows File Server.
Example:
Windows Application
↓
FSx for Windows File Server
↓
SMB File Share

Simple ga gurthupettuko:
Windows File Share = FSx for Windows

---

26. Which AWS service should you use to migrate on-premises files to AWS?
Answer:
AWS DataSync is commonly used to transfer large amounts of data between on-premises storage and AWS storage. It automates the transfer process and supports recurring transfers as well. It is useful for migrations, backups, and synchronization.
For example:
Suppose our company has files stored on an on-premises NFS server.
We can use DataSync to transfer those files to S3 or EFS.
Example:
On-Premises NFS
↓
DataSync
↓
S3 / EFS
Simple ga gurthupettuko:
On-Prem → AWS Data = DataSync

---

27. Which AWS service should you use when an existing application requires SFTP?
Answer:
AWS Transfer Family is commonly used when users or applications need managed SFTP, FTPS, FTP, or AS2 file transfer. It removes the need to maintain our own file-transfer server. The transferred files can be stored in supported AWS storage services.
For example:
Suppose an external partner sends daily files using SFTP.
We can create an AWS Transfer Family SFTP server and store the received files in S3.
Example:
External Partner
↓
SFTP
↓
Transfer Family
↓
S3
Simple ga gurthupettuko:
SFTP = Transfer Family

---

28. Which AWS service is useful for hybrid storage?
Answer:
AWS Storage Gateway is useful when applications are running on-premises but need access to AWS storage. It provides a bridge between on-premises environments and AWS storage services. This helps organizations gradually integrate cloud storage without immediately moving everything.
For example:
Suppose a company has an existing data center and wants to use S3 for additional storage.
Storage Gateway can provide a connection between the on-premises environment and AWS storage.
Example:
On-Premises
↓
Storage Gateway
↓
AWS
↓
S3
Simple ga gurthupettuko:
Hybrid Storage = Storage Gateway

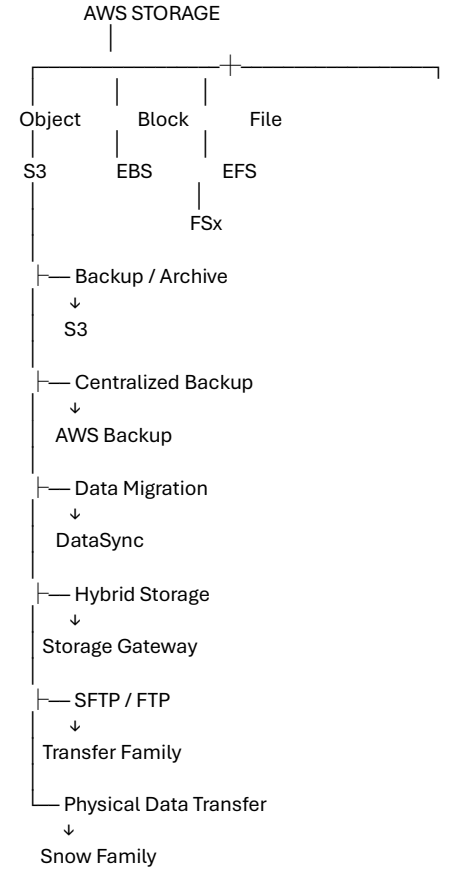
---

29. What is the easiest way to remember AWS storage services?
Answer:
We can remember AWS storage services based on the type of storage or the problem they solve. S3 is for objects, EBS is for EC2 block storage, EFS is for shared files, FSx is for specialized file systems, and DataSync is for moving data. Storage Gateway is mainly for hybrid storage and AWS Backup is for centralized backup management.
For example:
If an interviewer asks where you would store an image, think S3.
If they ask about an EC2 disk, think EBS, and if they ask about shared files, think EFS.
Example:
S3 → Object Storage
EBS → EC2 Block Storage
EFS → Shared File Storage
FSx → Specialized File System
DataSync → Data Transfer
Gateway → Hybrid Storage

Backup → Backup Management
Snow Family → Physical Data Transfer
Simple ga gurthupettuko:
S3 → EBS → EFS → FSx → Gateway → DataSync → Backup → Snow

---

30. Complete AWS Storage Decision Map
Answer:
The best AWS storage service depends on how the application wants to store and access data. We first identify whether the requirement is object, block, file, backup, hybrid, or data-transfer related. Then we select the AWS service designed for that requirement.
For example:
If our application stores images, we choose S3.
If EC2 needs a disk, we choose EBS, and if multiple servers need shared files, we choose EFS.
Example:



Simple ga gurthupettuko:
S3 = Objects | EBS = Disk | EFS = Shared Files | FSx = Special Files | DataSync = Move Data | Gateway = Hybrid | Backup = Protect | Snow = Physical Transfer